SYSCCA 2.0

Geospa - Installation Guide

VERSION 0.2

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1. Pendahuluan

Dokumen ini dibuat untuk memberikan panduan komprehensif mengenai bagaimana mempersiapkan, menguji, dan melakukan instalasi serta *deployment* aplikasi mendawai Geospatial.

2. Prasyarat Umum Pengetahuan

Untuk dapat mengikuti panduan instalasi dan *setup* sistem Asterisk, Issabel, serta layanan aplikasi geospasial ini dengan baik, Anda diwajibkan harus familier dan mengerti tentang hal-hal berikut:

- Linux OS
- Sistem Jaringan

3. Local Deployment

3.1. Prasyarat

- Node.js (v18.x)
- npm (v9.x)
- Docker Desktop
- Git
- Google Maps API Key (Maps JavaScript API; Places API; Geocoding API; Maps Static API; Street View Static API;)

3.2. Kloning Repositori

Buka terminal (Powershell/Bash) dan jalankan perintah berikut untuk mengambil source code:

```
git clone https://github.com/rizkimalik-webdev/mendawai-syscca-geospatial.git
cd mendawai-syscca-geospatial
cp .env.example .env
```

```
PS C:\Users\MxK\Desktop\test\geomap> git clone https://github.com/rizkimalik-webdev/mendawai-syscca-geospatial.git
Cloning into 'mendawai-syscca-geospatial'...
remote: Enumerating objects: 653, done.
PS C:\Users\MxK\Desktop\test\geomap> git clone https://github.com/rizkimalik-webdev/mendawai-syscca-geospatial.git
alik-webdev/mendawai-syscca-geospatial.git
Cloning into 'mendawai-syscca-geospatial'...
alik-webdev/mendawai-syscca-geospatial.git
Cloning into 'mendawai-syscca-geospatial'...
remote: Enumerating objects: 653, done.
remote: Counting objects: 100% (653/653), done.
remote: Compressing objects: 100% (237/237), done.
remote: Total 653 (delta 396), reused 653 (delta 396), pack-reused 0 (from 0)
Receiving objects: 100% (653/653), 310.81 KiB | 1.12 MiB/s, done.
Resolving deltas: 100% (396/396), done.
PS C:\Users\MxK\Desktop\test\geomap> cd mendawai-syscca-geospatial
PS C:\Users\MxK\Desktop\test\geomap\mendawai-syscca-geospatial> cp .env.example .env
```

3.3. Persiapan Database & Konfigurasi (.env)

Buka .env dan isi database url:

Isi Database URL dengan url database anda:

- BACKEND_DATABASE_URL=postgresql://user:password@host:5432/backend_db?schema=public&connection_limit=10
- GATEWAY_DATABASE_URL=postgresql://user:password@host:5432/gateway_db?schema=public&connection_limit=10

Contoh menjadi:

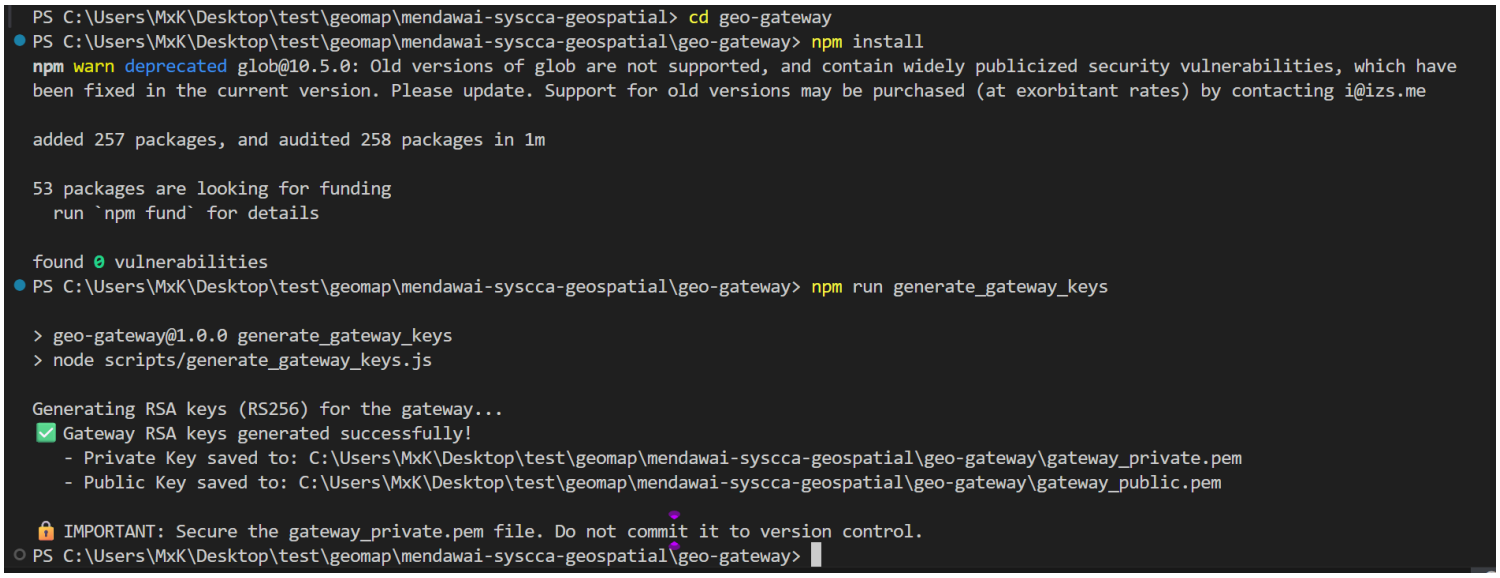
- BACKEND_DATABASE_URL=postgresql://user123:password1!@11.15.141.123:5432/backend_db?schema=public&connection_limit=16
- GATEWAY_DATABASE_URL=postgresql://user123:password1!@11.15.141.123:5432/gateway_db?schema=public&connection_limit=8

3.4. Membuat Kunci Keamanan

Aplikasi membutuhkan sepasang kunci (Public/Private Key) untuk sistem keamanan token (JWT).

Jalankan perintah:

- cd geo-gateway
- npm install
- npm run generate_gateway_keys



3.5. Pindahkan Kunci ke Folder keys

Jalankan perintah:

- mkdir -p ../keys
- mv gateway_private.pem ../keys/
- mv gateway_public.pem ../keys/
- cd ..

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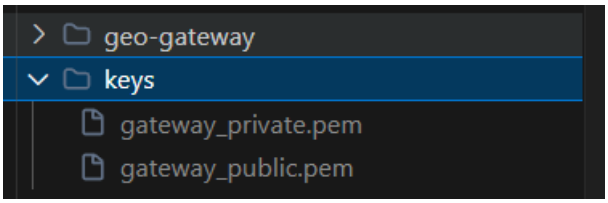
```
PS C:\Users\MxK\Desktop\test\geomap\mendawai-syscca-geospatial\geo-gateway> mkdir -p ../keys


    Directory: C:\Users\MxK\Desktop\test\geomap\mendawai-syscca-geospatial


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----        08/04/2026     15.45                keys


PS C:\Users\MxK\Desktop\test\geomap\mendawai-syscca-geospatial\geo-gateway> mv gateway_private.pem ../keys/
PS C:\Users\MxK\Desktop\test\geomap\mendawai-syscca-geospatial\geo-gateway> mv gateway_public.pem ../keys/
PS C:\Users\MxK\Desktop\test\geomap\mendawai-syscca-geospatial\geo-gateway> cd ..
PS C:\Users\MxK\Desktop\test\geomap\mendawai-syscca-geospatial>
```

Pastikan kunci sudah berada di folder keys di bawah geo-gateway:



3.6. Registrasi Klien & Mendapatkan Client ID

Karena ini instalasi baru, kita harus mendaftarkan "Klien" agar mendapatkan ID unik.

Jalankan perintah:

```
docker compose up -d gateway backend
```

```
[+] Running 5/5
 ✓ backend                                    Built
 ✓ gateway                                    Built
 ✓ Network mendawai-syscca-geospatial_default Created
 ✓ Container mendawai-syscca-geospatial-gateway-1 Healthy
 ✓ Container mendawai-syscca-geospatial-backend-1 Started
```

Jalankan perintah:

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- docker compose exec gateway npm run register_client

```
PS C:\Users\MxK\Desktop\test\geomap\mendawai-syscca-geospatial> docker compose exec gateway npm run register_client

> geo-gateway@1.0.0 register_client
> node scripts/register_client.js

--- Client Registration (Interactive) ---
➡Masukkan nama klien (misal, react-frontend-app):local-testing
➡Masukkan origins yang diizinkan (pisahkan koma, misal, http://localhost:5173):http://localhost:14170
```

Masukkan nama klien, contoh: **local-testing**

Masukkan masukkan origins: **http://localhost:14170**

```
🔄 Menghasilkan key pair RS256...
✅ Key pair dihasilkan.
🔄 Menyimpan klien 'local-testing' ke database...

--- ✅ Registrasi Klien Berhasil! ---
Client Name: local-testing
Auth Method: client_credentials
Allowed Origins: http://localhost:14170

🔒 Kunci privat disimpan dengan aman di database.

🔑 **Client ID (untuk digunakan di .env frontend):**
   client_2d6899af-1270-4c90-b3f3-883e84fecb18
------------------------------------------------
PS C:\Users\MxK\Desktop\test\geomap\mendawai-syscca-geospatial>
```

Salin dan simpan Client ID (client_2d6899af-1270-4c90-b3f3-883e84fecb18) untuk digunakan nanti.

3.7. Konfigurasi .env

Lengkapi konfigurasi URL:

- GATEWAY_URL=http://localhost

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```
RESOURCE_SERVER_URL=http://backend:3001
AI_SERVICE_URL=https://localhost

# Ubah konfigurasi di atas menjadi:
GATEWAY_URL=http://localhost: 13000
RESOURCE_SERVER_URL=http://backend:3001
# Isi dengan url Geospa AI Service anda
AI_SERVICE_URL=http://39.97.87.194:3008
```

Masukkan Client ID dan Google Maps API Key:

```
VITE_GOOGLE_MAPS_API_KEY=your_google_maps_key
VITE_CLIENT_ID=your_client_id

# Ubah konfigurasi di atas menjadi:
# Isi dengan google maps api key anda
VITE_GOOGLE_MAPS_API_KEY=AIzaSy-ContohAPIKeyGoogleMapsDev000
# Masukkan Client ID sebelumnya
VITE_CLIENT_ID=client_2d6899af-1270-4c90-b3f3-883e84fecb18
```

Contoh hasil akhir konfigurasi:

```
# --- General Configuration ---
NODE_ENV=production
# Container host port bindings (Nginx Global handles public-facing ports)
FRONTEND_PORT=14170
BACKEND_PORT=13001
GATEWAY_PORT=13000

# --- Database Connection ---
# Format:
postgresql://USER:PASSWORD@HOST:PORT/DATABASE?schema=public
```

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```
BACKEND_DATABASE_URL=postgresql://user123:password1!@11.15.141.123:5432/backend_db?schema=public&connection_limit=16
GATEWAY_DATABASE_URL=postgresql://user123:password1!@11.15.141.123:5432/gateway_db?schema=public&connection_limit=8

# --- URLs and Ports ---
# These are used by the backend/gateway to talk to each other or the world
GATEWAY_URL=http://localhost:13000
RESOURCE_SERVER_URL=http://backend:3001
AI_SERVICE_URL=http://39.97.87.194:3008

# --- Frontend Build Variables ---
# Required for the React app build process
VITE_GOOGLE_MAPS_API_KEY=AIzaSy-ContohAPIKeyGoogleMapsDev000
VITE_CLIENT_ID=client_2d6899af-1270-4c90-b3f3-883e84fecb18
VITE_APP_BASE_URL=/

# --- Keys ---
# Paths inside the container
GATEWAY_PRIVATE_KEY_PATH=/app/keys/gateway_private.pem
GATEWAY_PUBLIC_KEY_PATH=/app/keys/gateway_public.pem
```

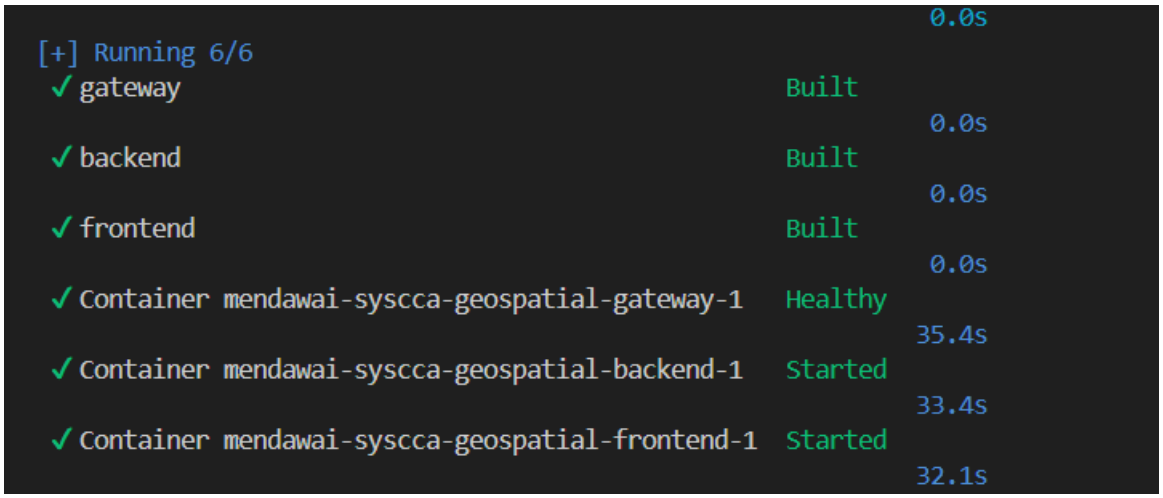
Anda bisa melihat konfigurasi sebelum sesudahnya DI SINI

3.8. Jalankan deployment

Jalankan perintah:

```
docker compose up -d --build
```

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3.9. Verifikasi

Buka url http://localhost:14170/ untuk membuka aplikasi geospa.

4. Production Deployment

4.1. Prasyarat

- Docker (v20.x)
- Docker Compose (v2.x)
- Git
- Nginx
- OpenSSL
- Google Maps API Key (Maps JavaScript API; Places API; Geocoding API; Maps Static API; Street View Static API;)

4.2. Instalasi Docker

Login ke dalam server/vps anda:

```
ssh icon@10.14.151.119

# Masukkan password akun jika diminta password.
```

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```
PS C:\Users\MxK\Desktop\Kiero Project\App\map-batas-wilayah\reactJS-webapp> ssh icon@10.14.151.119
icon@10.14.151.119's password:
```

Tampilan jika berhasil login:

```
Last login: Wed Apr  8 17:16:35 2026 from 10.14.92.26
icon@geospatial-dev:~$
```

Pastikan sistem Anda mutakhir dan instal paket bantuan:

- sudo apt update && sudo apt upgrade -y
- sudo apt install -y ca-certificates curl gnupg lsb-release

Masukkan password jika diminta

Cek docker dengan perintah:

- docker --version
- docker compose version

Jika tidak berhasil atau versi di bawah Prasyarat, silakan install docker dengan perintah:

Tambahkan GPG Key resmi

- sudo install -m 0755 -d /etc/apt/keyrings
- curl -fsSL https://download.docker.com/linux/ubuntu/gpg | sudo gpg --dearmor -o /etc/apt/keyrings/docker.gpg
- sudo chmod a+r /etc/apt/keyrings/docker.gpg

Daftarkan repositori

- echo "deb [arch=$(dpkg --print-architecture) signed-by=/etc/apt/keyrings/docker.gpg]

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https://download.docker.com/linux/ubuntu $(. /etc/os-release && echo "$VERSION_CODENAME") stable" | sudo tee /etc/apt/sources.list.d/docker.list > /dev/null

Instalasi Docker & Compose Plugin

- sudo apt update
- sudo apt install -y docker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-compose-plugin

Pengaturan Hak Akses
Ganti 'icon' dengan username Anda jika berbeda

- sudo usermod -aG docker icon

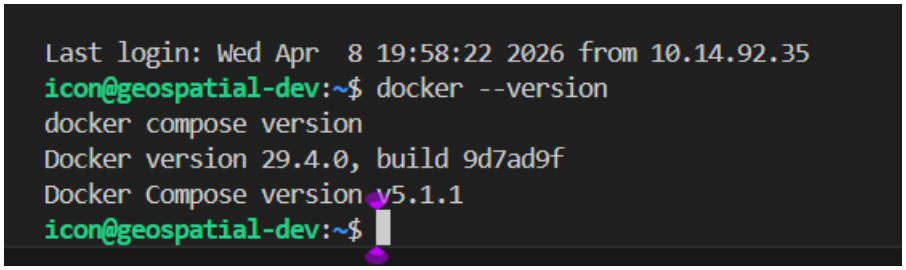
(PENTING: Anda harus logout dan login kembali agar perubahan ini berlaku)

Aktivasi & Verifikasi docker:

- sudo systemctl enable docker
- sudo systemctl start docker

Cek apakah sudah terinstal dengan benar

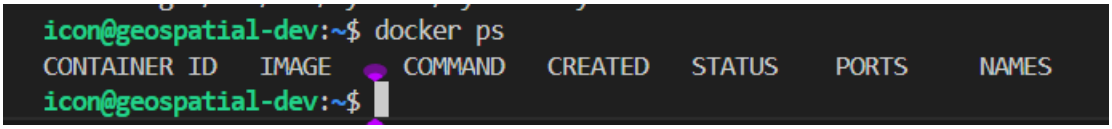
- docker --version
- docker compose version



Verifikasi Izin Docker:

- docker ps

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4.3. Instalasi Nginx

Cek nginx dengan perintah:

- nginx -v
- systemctl status nginx

Jika tidak berhasil atau versi di bawah Prasyarat, silakan install nginx dengan perintah:

- sudo apt update

Instal Nginx

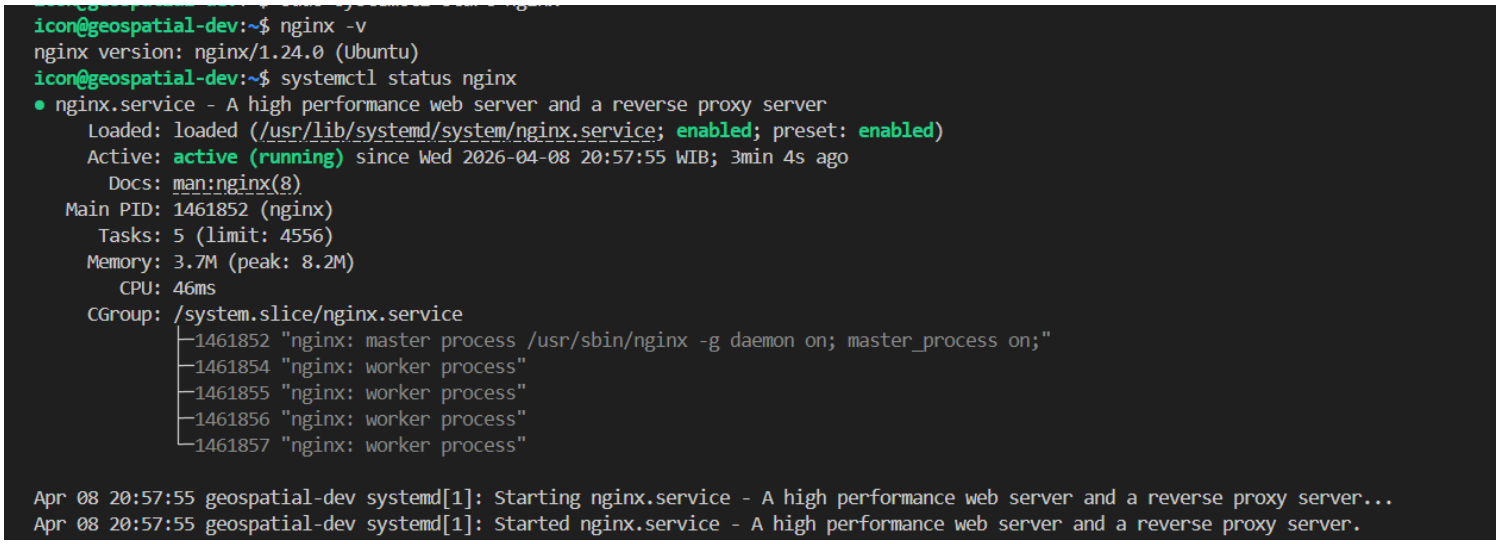
- sudo apt install -y nginx

Pastikan layanan berjalan

- sudo systemctl enable nginx
- sudo systemctl start nginx

Verifikasi instalasi nginx dengan perintah:

- nginx -v
- systemctl status nginx



Optimasi File nginx.conf:

```
# Backup konfigurasi nginx yang ada
sudo cp /etc/nginx/nginx.conf /etc/nginx/nginx.conf.bak

# Buat SSL DH Param
sudo openssl dhparam -out /etc/nginx/dhparam.pem 2048

# Update nginx.conf dengan konfigurasi ini
sudo bash -c 'cat << EOF > /etc/nginx/nginx.conf
user www-data;
worker_processes auto;
worker_rlimit_nofile 65535;
pid /run/nginx.pid;
include /etc/nginx/modules-enabled/*.conf;

events {
    worker_connections 4096;
    multi_accept on;
    use epoll;
}

http {
```

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```
    # ——— Security Hardening ———————————————————————————————————————
    server_tokens off;
    add_header X-Content-Type-Options "nosniff" always;
    charset utf-8;

    # ——— MIME & Optimization ———————————————————————————————————————
    include      /etc/nginx/mime.types;
    default_type  application/octet-stream;

    # ——— Buffer Optimization ———————————————————————————————————————
    client_body_buffer_size    16K;
    client_header_buffer_size  1k;
    client_max_body_size       50M;
    large_client_header_buffers 4 8k;

    # ——— Timeouts ——————————————————————————————————————————————————
    client_body_timeout   12;
    client_header_timeout 12;
    keepalive_timeout     65;
    send_timeout          10;

    # ——— SSL Hardening (Mozilla Intermediate) ——————————————————————
    ssl_protocols TLSv1.2 TLSv1.3;
    ssl_prefer_server_ciphers on;
    ssl_session_cache   shared:SSL:10m;
    ssl_session_timeout 1d;
    ssl_session_tickets off;

    # DH Parameters untuk Forward Secrecy
    ssl_dhparam /etc/nginx/dhparam.pem;

    # ——— Gzip Compression ——————————————————————————————————————————
```

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```
    gzip on;
    gzip_vary on;
    gzip_proxied any;
    gzip_comp_level 6;
    gzip_min_length 256;
    gzip_types
        text/plain
        text/css
        text/xml
        text/javascript
        application/json
        application/javascript
        application/xml
        image/svg+xml
        font/woff2;

    # ——— Logging ———————————————————————————————————————
————————————————————————

    access_log /var/log/nginx/access.log;
    error_log  /var/log/nginx/error.log;

    # ——— Includes ——————————————————————————————————————
————————————————————

    include /etc/nginx/conf.d/*.conf;
    include /etc/nginx/sites-enabled/*;
}
EOF'
```

Optimasi konfigurasi proxy param. Buka dan edit proxy_params:

```
sudo nano /etc/nginx/proxy_params
```

Isi proxy_params dengan konfigurasi berikut:

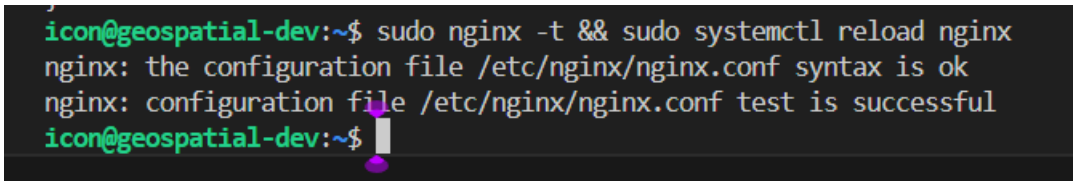
```
proxy_set_header Host            $http_host;
proxy_set_header X-Real-IP       $remote_addr;
proxy_set_header X-Forwarded-For   $proxy_add_x_forwarded_for;
```

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```
proxy_set_header X-Forwarded-Proto $scheme;
proxy_set_header X-Forwarded-Host  $host;
proxy_set_header X-Forwarded-Port  $server_port;
```

Tes dan reload konfigurasi:

```
sudo nginx -t && sudo systemctl reload nginx
```



4.4. Instalasi Git & OpenSSL

Cek Git & OpenSSL dengan perintah:

```
# Cek Git
git --version

# Cek OpenSSL
openssl version
```

Jika tidak berhasil atau versi di bawah Prasyarat, silakan install nginx dengan perintah:

```
sudo apt install -y git openssl
```

Verifikasi instalasi Git & OpenSSL dengan perintah:

```
nginx -v
systemctl status nginx
```

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```
git version 2.43.0
OpenSSL 3.0.13 30 Jan 2024 (Library: OpenSSL 3.0.13 30 Jan 2024)
icon@geospatial-dev:~$
```

4.5. Konfigurasi Firewall

Cek status firewall:

Cek firewall

- sudo ufw status

Cek port yang sedang digunakan

- sudo ss -tlnp

```
icon@geospatial-dev:~$ sudo ufw status
[sudo] password for icon:
Status: inactive
icon@geospatial-dev:~$ sudo ss -tlnp
State    Recv-Q   Send-Q     Local Address:Port     Peer Address:Port  Process
LISTEN   0        4096           127.0.0.54:53          0.0.0.0:*      users:(("systemd-resolve",pid=53461,fd=17))
LISTEN   0        511              0.0.0.0:80          0.0.0.0:*      users:(("nginx",pid=1461857,fd=5),("nginx",pid=1461856,fd=5),("nginx",pid=1
461855,fd=5),("nginx",pid=1461854,fd=5),("nginx",pid=1461852,fd=5))
LISTEN   0        4096       127.0.0.53%lo:53          0.0.0.0:*      users:(("systemd-resolve",pid=53461,fd=15))
LISTEN   0        511                 [::]:80             [::]:*      users:(("nginx",pid=1461857,fd=6),("nginx",pid=1461856,fd=6),("nginx",pid=1
461855,fd=6),("nginx",pid=1461854,fd=6),("nginx",pid=1461852,fd=6))
LISTEN   0        4096                  *:22                *:*      users:(("sshd",pid=15132,fd=3),("systemd",pid=1,fd=83))
```

Lihat Local Address:Port. Daftarkan semua port kecuali yang localhost(127.0.0.x):

1. Izinkan SSH (WAJIB PERTAMA, agar tidak terkunci dari server!)

- sudo ufw allow OpenSSH

2. Izinkan rute Web (untuk Nginx Reverse Proxy)

- sudo ufw allow 80/tcp
- sudo ufw allow 443/tcp

3. Izinkan port publik aplikasi Geomap

- sudo ufw allow 4170/tcp
- sudo ufw allow 3000/tcp

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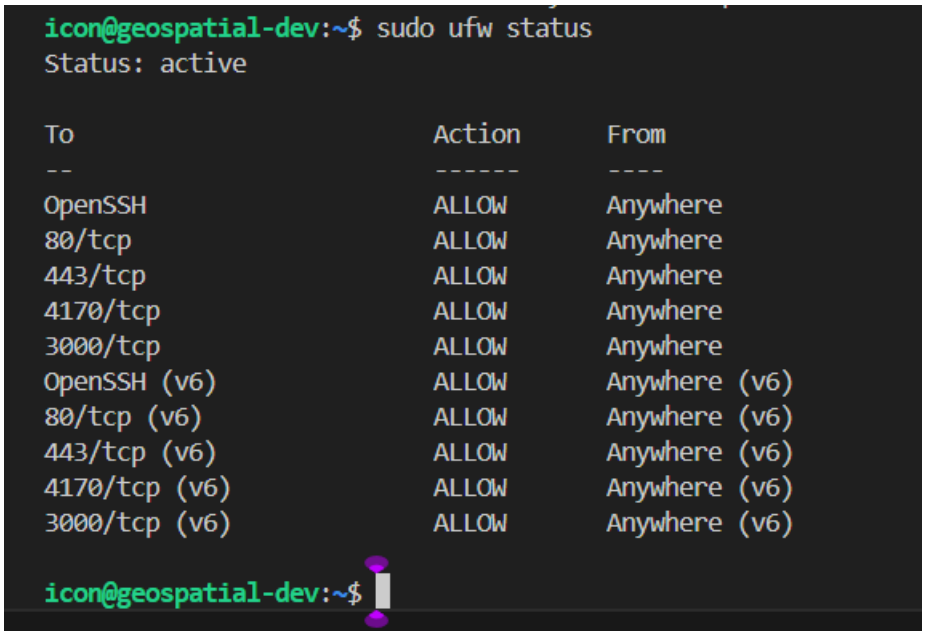
Aktifkan firewal dan verifikasi

Aktifkan Firewall (ketik 'y' saat diminta konfirmasi)

- sudo ufw enable

Verifikasi

- sudo ufw status



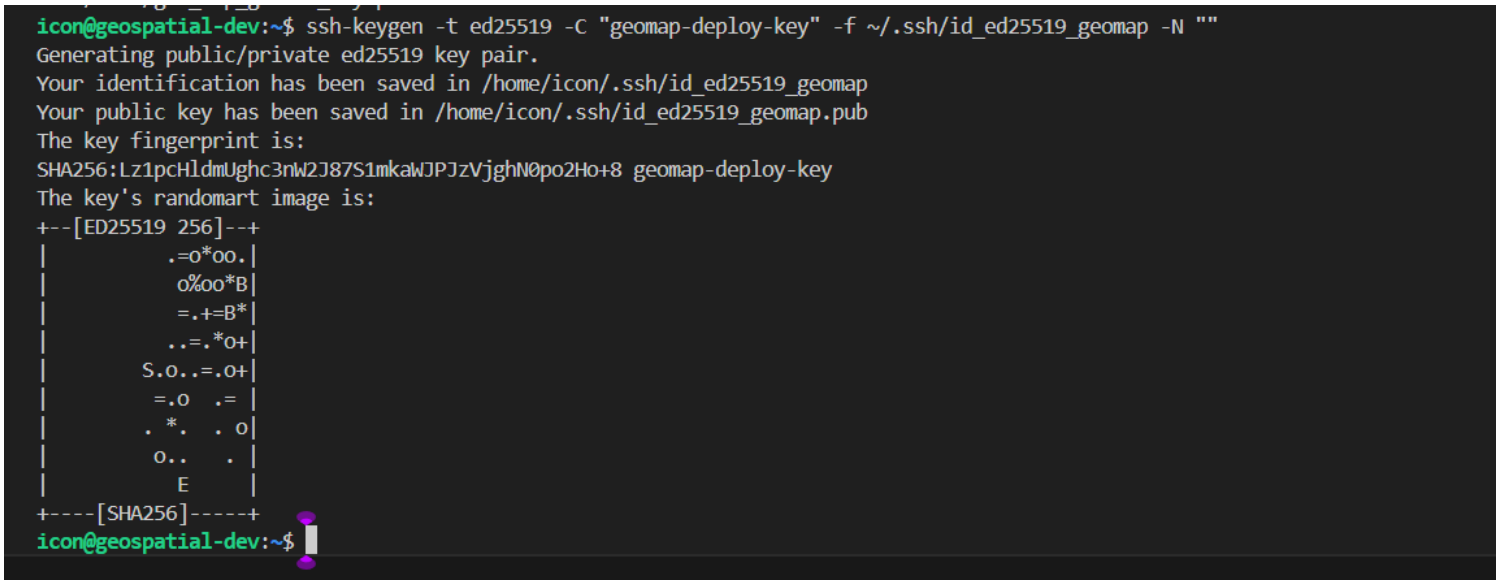
4.6. Kloning Repositori

Buat SSH key khusus proyek Geomap. Kita akan menggunakan algoritma ed25519.

Jalankan perintah:

- ssh-keygen -t ed25519 -C "geomap-deploy-key" -f ~/.ssh/id_ed25519_geomap -N ""

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Daftarkan "Alias" Khusus di SSH Config dengan perintah:

```
# Buat file config jika belum ada
touch ~/.ssh/config

# Tambahkan konfigurasi host alias
cat << 'EOF' >> ~/.ssh/config
Host github.com-geomap
 HostName github.com
 User git
 IdentityFile ~/.ssh/id_ed25519_geomap
 IdentitiesOnly yes
EOF
```

Verifikasi konfigurasi SSH:

```
cat ~/.ssh/config
```

Ambil public key untuk github:

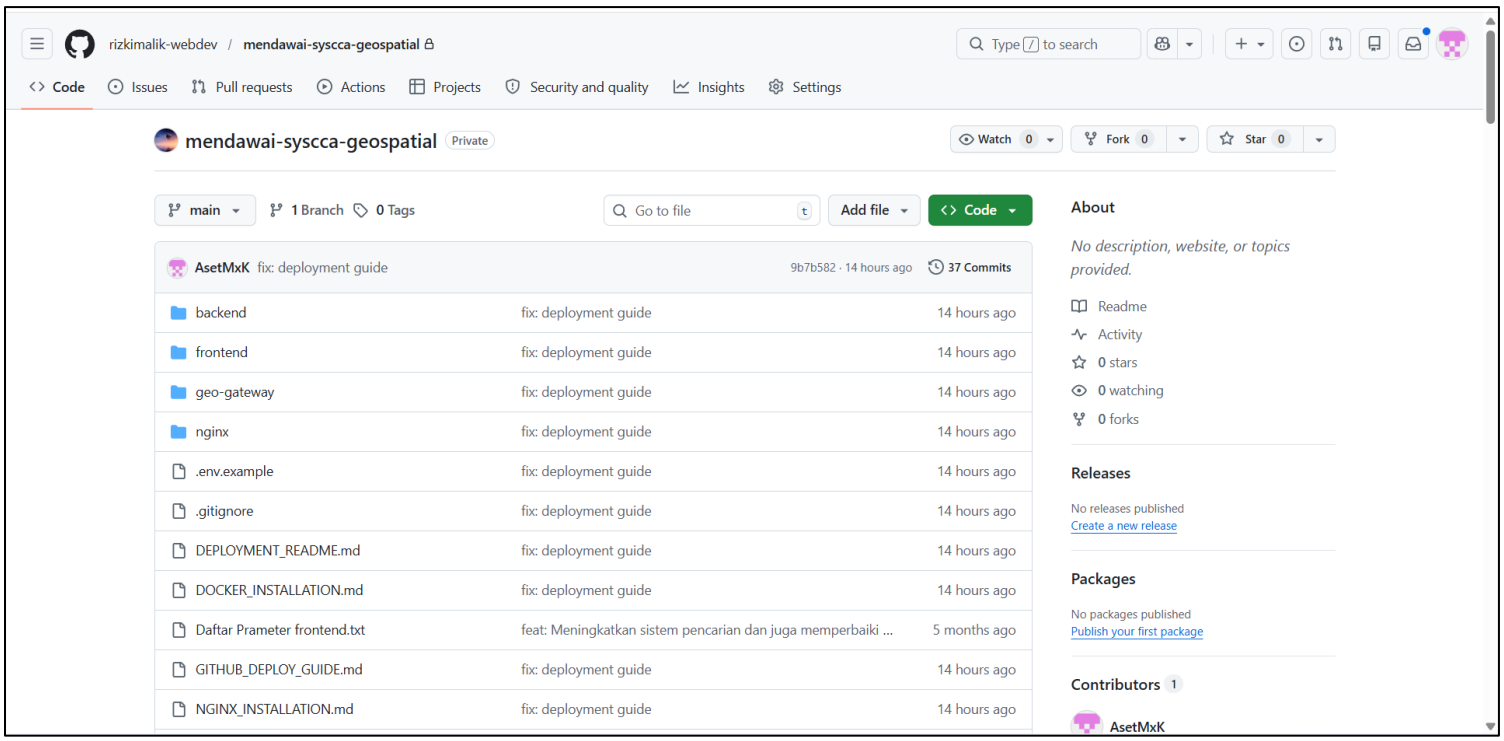
```
cat ~/.ssh/id_ed25519_geomap.pub
```

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```
# Salin dan simpan output-nya (dimulai dari ssh-ed25519)
```

Daftarkan di GitHub Repository:

1. Buka browser komputer Anda dan akses **Repository GitHub** geospa Anda.

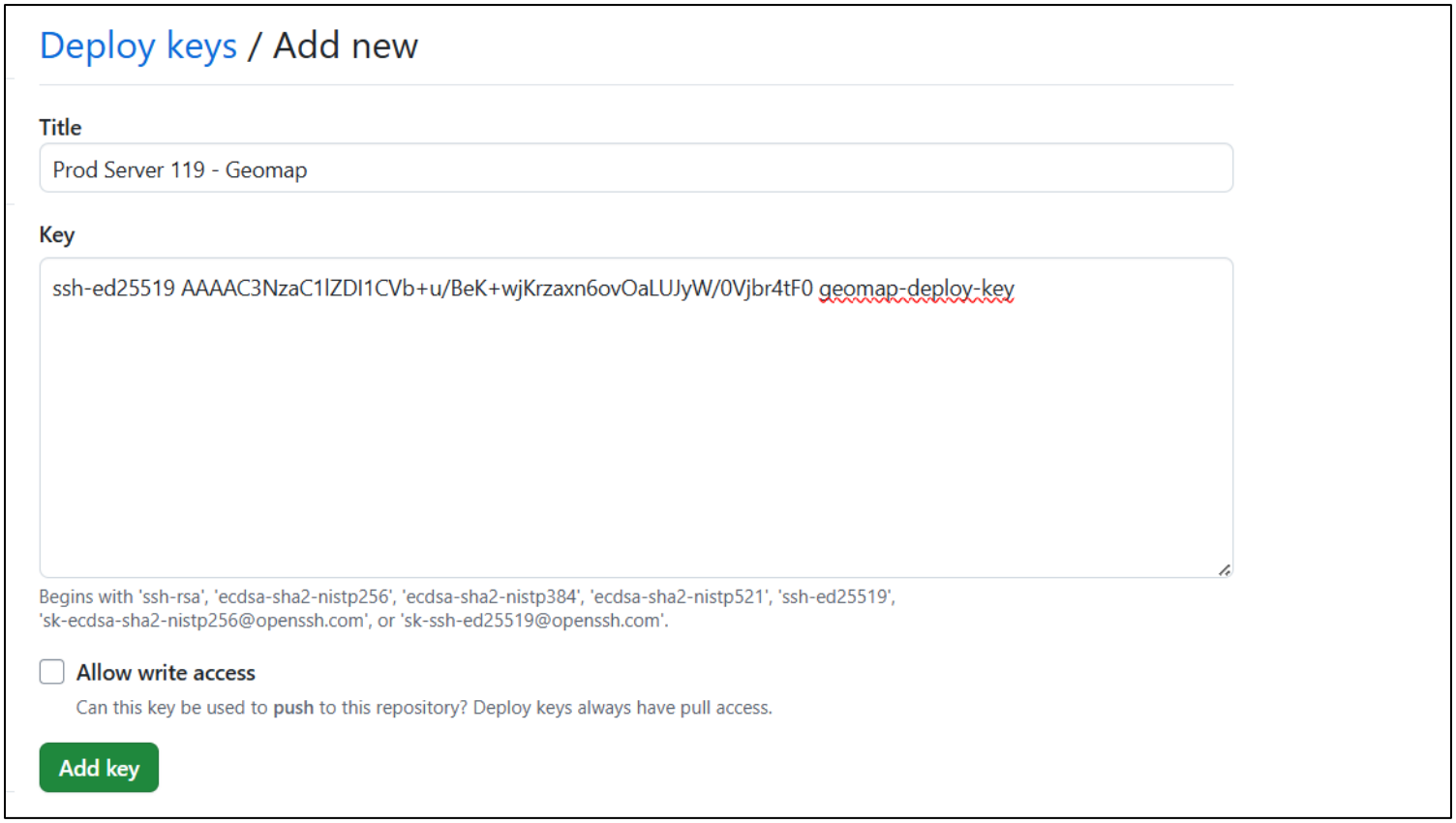


2. Klik tab **Settings** (Pengaturan repositori).
3. Di panel kiri, masuk ke menu **Deploy keys** (dibawah bagian *Security*).

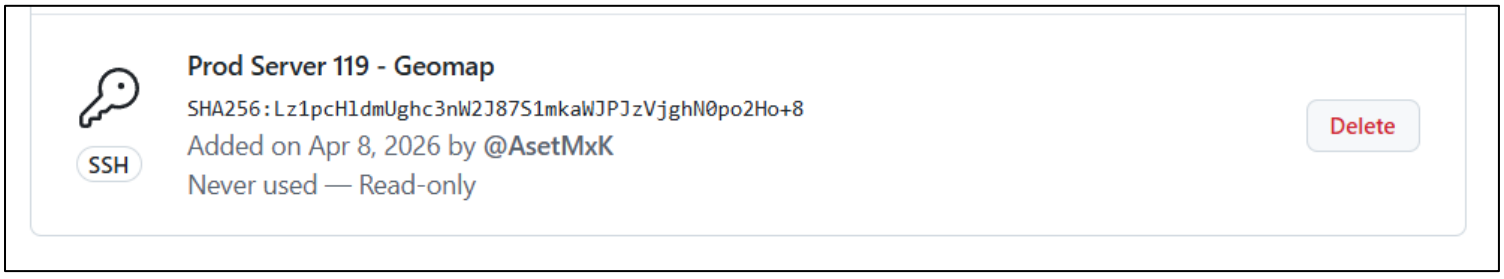


4. Klik tombol **Add deploy key**.

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5. Isi formulir:
    - **Title**: Prod Server 119 - Geomap
    - **Key**: *Paste* (tempelkan) *Public Key* yang sudah disalin di Langkah 3.
    - **Allow write access**: **JANGAN DICENTANG** (Biarkan Read-only demi keamanan server).
6. Klik **Add key**.



Uji koneksi:

```
ssh -T git@github.com-geomap
```

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```
icon@geospatial-dev:~$ ssh -T git@github.com-geomap
Hi AsetMxK/Geomap! You've successfully authenticated, but GitHub does not provide shell access.
icon@geospatial-dev:~$
```

Kloning repositori:

- cd /home/icon

PASTIKAN GANTI TULISAN USERNAME DI BAWAH DENGAN AKUN GITHUB ANDA DAN REPOSITORI DENGAN NAMA REPOSITORI ANDA

- git clone git@github.com-geomap:USERNAME/REPOSITORI.git geomap

```
icon@geospatial-dev:~$ git clone git@github.com-geomap:AsetMxK/Geomap.git geomap
Cloning into 'geomap'...
remote: Enumerating objects: 669, done.
remote: Counting objects: 100% (669/669), done.
remote: Compressing objects: 100% (439/439), done.
remote: Total 669 (delta 410), reused 463 (delta 210), pack-reused 0 (from 0)
Receiving objects: 100% (669/669), 869.87 KiB | 800.00 KiB/s, done.
Resolving deltas: 100% (410/410), done.
icon@geospatial-dev:~$
```

4.7. Setup proyek

Buat folder keys & logs:

- cd /home/icon/geomap

Pastikan folder keys dan logs tersedia

- mkdir -p keys
- mkdir -p logs

Buat paired keys:

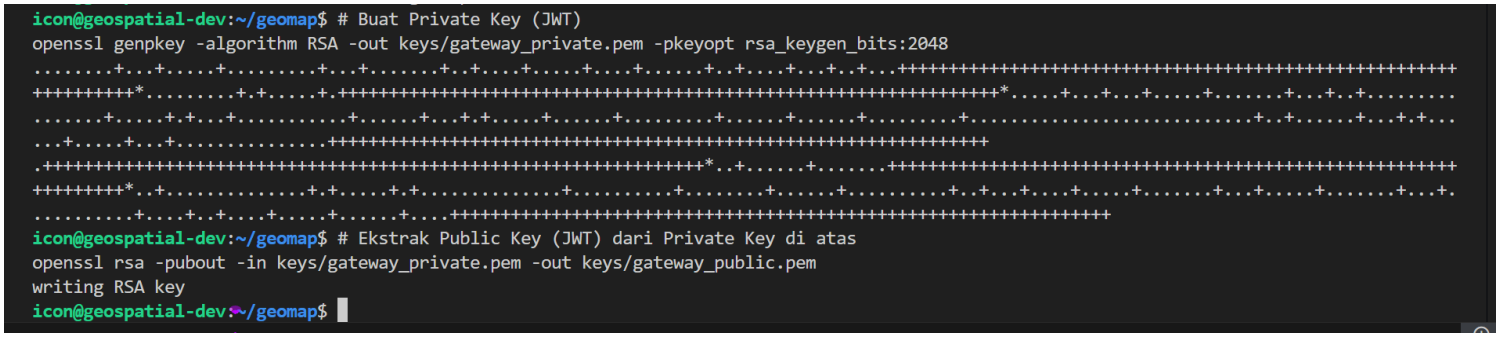
Buat Private Key (JWT)

- openssl genpkey -algorithm RSA -out keys/gateway_private.pem -pkeyopt rsa_keygen_bits:2048

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Ekstrak Public Key (JWT) dari Private Key di atas

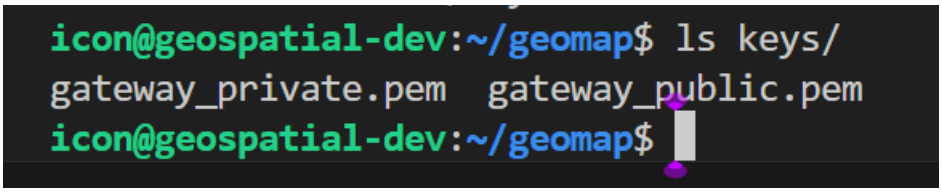
- openssl rsa -pubout -in keys/gateway_private.pem -out keys/gateway_public.pem



Verifikasi keys:

Cek isi folder keys

- ls keys/



Membuat SSL (selfsigned) untuk https:

- cat << 'EOF' > generate_ssl.sh

```
#!/bin/bash
# 🔐 Optimized Self-Signed SSL Generator with SAN support

# 1. Deteksi IP Otomatis atau Gunakan Argumen
# Jika menjalankan ./generate_ssl.sh 1.2.3.4 maka akan pakai IP tersebut.
# Jika tidak, akan mendeteksi IP utama server.
DEFAULT_IP=$(hostname -I | awk '{print $1}')
SERVER_IP=${1:-$DEFAULT_IP}
```

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```
KEY_DIR="./keys/ssl"
CONFIG_FILE="./keys/ssl/openssl.cnf"
mkdir -p $KEY_DIR

echo "🔍 Target IP: $SERVER_IP"

# 2. Buat file konfigurasi OpenSSL sementara
cat << EOC > $CONFIG_FILE
[req]
default_bits       = 2048
distinguished_name = req_distinguished_name
x509_extensions    = v3_req
prompt             = no

[req_distinguished_name]
C  = ID
ST = DKI Jakarta
L  = Jakarta
O  = Kiero Project
OU = IT Department
CN = $SERVER_IP

[v3_req]
keyUsage = nonRepudiation, digitalSignature, keyEncipherment
extendedKeyUsage = serverAuth
subjectAltName = @alt_names

[alt_names]
IP.1 = $SERVER_IP
IP.2 = 127.0.0.1
DNS.1 = localhost
EOC

echo "🔐 Generating SSL Certificate (Valid for 10 years)..."

openssl req -x509 -nodes -days 3650 -newkey rsa:2048 \
  -keyout "$KEY_DIR/nginx.key" \
  -out "$KEY_DIR/nginx.crt" \
  -config $CONFIG_FILE
```

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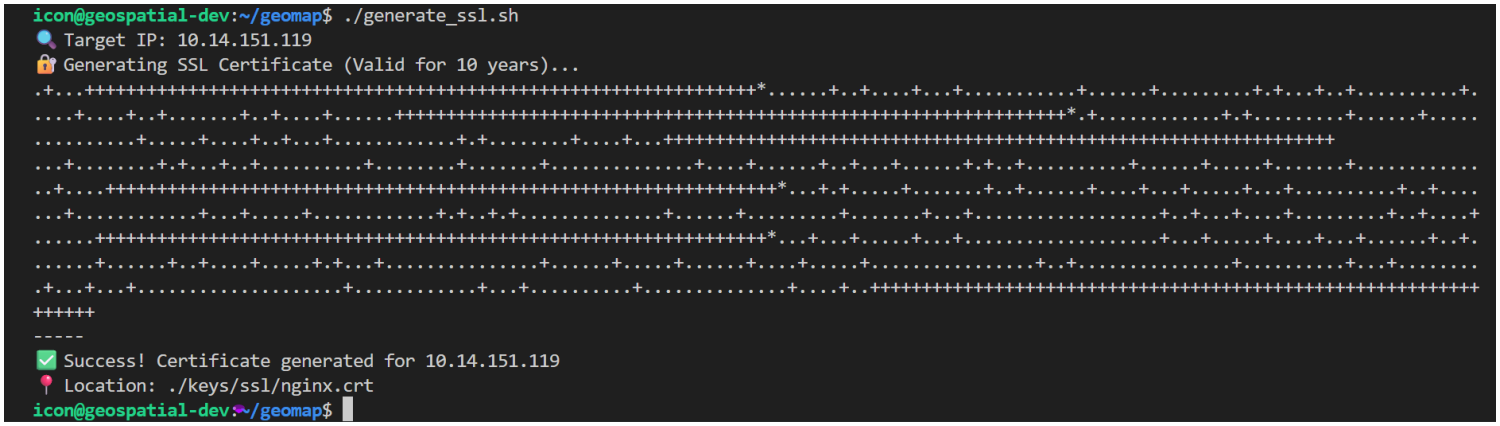
```
# Hapus file konf sementara
rm $CONFIG_FILE

echo "☑ Success! Certificate generated for $SERVER_IP"
echo " 📍 Location: $KEY_DIR/nginx.crt"
EOF
```

Jalankan skrip:

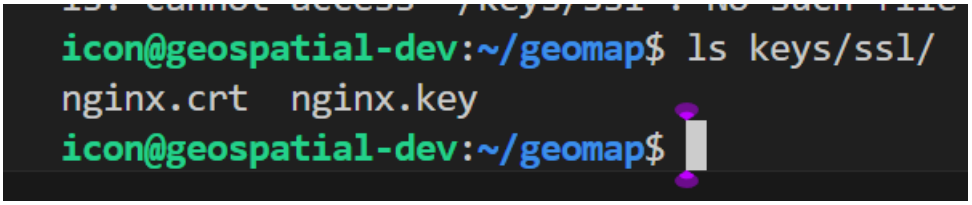
```
# Berikan izin eksekusi
chmod +x generate_ssl.sh

# Eksekusi pembuat sertifikat 10 tahun
./generate_ssl.sh
```



Verifikasi sertifikat SSL:

```
ls keys/ssl/
```



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Salin SSL ke direktori host Nginx:

```
sudo cp keys/ssl/nginx.crt /etc/ssl/certs/geomap-selfsigned.crt
sudo cp keys/ssl/nginx.key /etc/ssl/private/geomap-selfsigned.key
```

Membuat auto-renewal SSL:

```
cat << 'EOF' > renew_ssl.sh
#!/bin/bash
# 🔄 SSL Auto-Renewal Script for Self-Signed Certs

PROJECT_DIR="/home/icon/geomap"
CERT_FILE="$PROJECT_DIR/keys/ssl/nginx.crt"
DAYS_BEFORE_EXPIRY=30

echo "📆 Checking SSL certificate expiry..."

# 1. Cek apakah file sertifikat ada
if [ ! -f "$CERT_FILE" ]; then
  echo "❌ Certificate not found at $CERT_FILE. Generating new one..."
  cd $PROJECT_DIR && ./generate_ssl.sh
else
  # 2. Cek sisa umur sertifikat dalam detik
  # openssl x509 -checkend akan return 0 jika cert masih valid sampai N detik ke depan
  EXPIRY_SECONDS=$((DAYS_BEFORE_EXPIRY * 24 * 3600))

  if openssl x509 -checkend $EXPIRY_SECONDS -noout -in "$CERT_FILE"; then
    echo "☑️ Certificate is still valid for more than $DAYS_BEFORE_EXPIRY days. No renewal needed."
    exit 0
  else
    echo "⚠️ Certificate is expiring soon (less than $DAYS_BEFORE_EXPIRY days). Renewing..."
    cd $PROJECT_DIR && ./generate_ssl.sh
  fi
```

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```
fi

# 3. Salin ke direktori sistem Nginx (Requires sudo)
echo "📂 Copying new certificates to system directory..."
sudo cp "$PROJECT_DIR/keys/ssl/nginx.crt" /etc/ssl/certs/geomap-selfsigned.crt
sudo cp "$PROJECT_DIR/keys/ssl/nginx.key" /etc/ssl/private/geomap-selfsigned.key

# 4. Reload Nginx
echo "🚀 Reloading Nginx..."
sudo systemctl reload nginx

echo "✨ Renewal process completed successfully."
EOF
```

Setup auto-renewal dengan cron job:

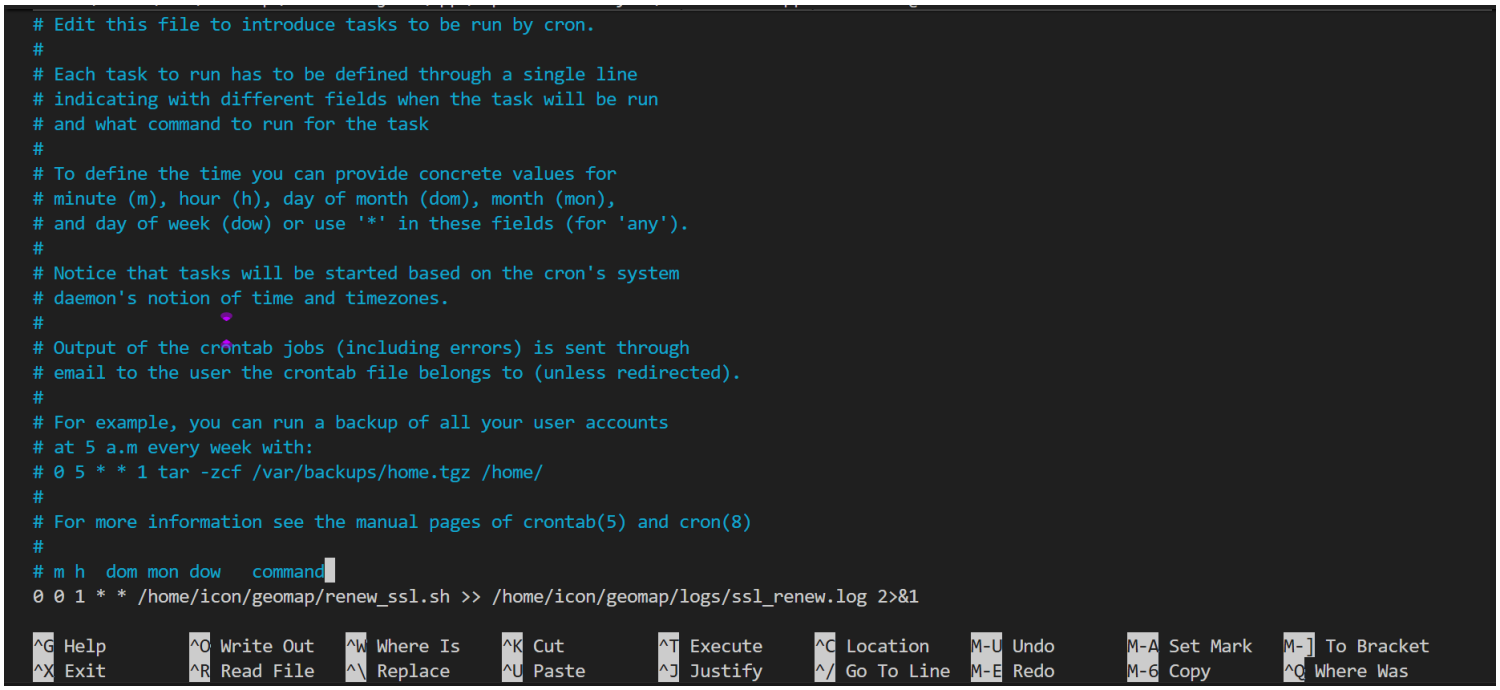
```
# Berikan izin eksekusi skrip auto-renewal SSL
chmod +x renew_ssl.sh

# Buka editor crontab
crontab -e
```

Tambahkan baris ini di paling bawah:

```
# menjalankan pengecekan setiap tanggal 1 jam 00:00
0 0 1 * * /home/icon/geomap/renew_ssl.sh >> /home/icon/geomap/logs/ssl_renew.log 2>&1
```

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Konfigurasi .env:

Salin .env.example

- cd /home/icon/geomap
- cp .env.example .env

Edit .env

- nano .env

Buka .env dan isi database url:

Isi Database URL dengan url database anda:

- BACKEND_DATABASE_URL=postgresql://user:password@host:5432/backend_db?schema=public&connection_limit=10
- GATEWAY_DATABASE_URL=postgresql://user:password@host:5432/gateway_db?schema=public&connection_limit=10

Contoh menjadi:

- BACKEND_DATABASE_URL=postgresql://user123:password1!@11.15.141.123:5432/backend_db?schema=public&connection_limit=16

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- GATEWAY_DATABASE_URL=postgresql://user123:password1!@11.15.141.123:5432/gateway_db?schema=public&connection_limit=8

Lengkapi konfigurasi URL:

- GATEWAY_URL=http://localhost
- RESOURCE_SERVER_URL=http://backend:3001
- AI_SERVICE_URL=https://localhost

Ubah konfigurasi di atas menjadi:

- GATEWAY_URL=https://10.14.151.119:3000
- RESOURCE_SERVER_URL=http://backend:3001

Isi dengan url Geospa AI Service anda

- AI_SERVICE_URL=http://39.97.87.194:3008

Simpan konfigurasi .env. Hasilnya contoh akan seperti di bawah:

--- General Configuration ---

- NODE_ENV=production
- # Container host port bindings (Nginx Global handles public-facing ports)
- FRONTEND_PORT=14170
- BACKEND_PORT=13001
- GATEWAY_PORT=13000
-
- # --- Database Connection ---
- # Format: postgresql://USER:PASSWORD@HOST:PORT/DATABASE?schema=public
- BACKEND_DATABASE_URL=postgresql://user123:password1!@11.15.141.123:5432/backend_db?schema=public&connection_limit=16
- GATEWAY_DATABASE_URL=postgresql://user123:password1!@11.15.141.123:5432/gateway_db?schema=public&connection_limit=8
-

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```
# --- URLs and Ports ---
# These are used by the backend/gateway to talk to each other or the world
GATEWAY_URL=https://10.14.151.119:3000
RESOURCE_SERVER_URL=http://backend:3001
AI_SERVICE_URL=http://39.97.87.194:3008

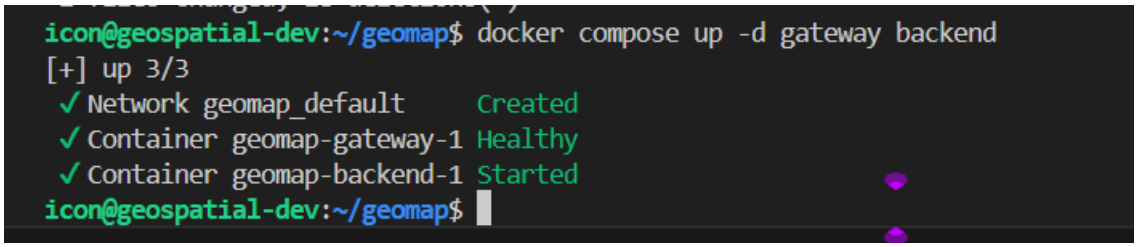
# --- Frontend Build Variables ---
# Required for the React app build process
VITE_GOOGLE_MAPS_API_KEY=your_google_maps_key
VITE_CLIENT_ID=your_client_id
VITE_APP_BASE_URL=/

# --- Keys ---
# Paths inside the container
GATEWAY_PRIVATE_KEY_PATH=/app/keys/gateway_private.pem
GATEWAY_PUBLIC_KEY_PATH=/app/keys/gateway_public.pem
```

Lihat perbedaan perubahannya DI SINI.

Dapatkan Client ID dengan perintah berikut:

```
cd /home/icon/geomap
docker compose up -d gateway backend
```



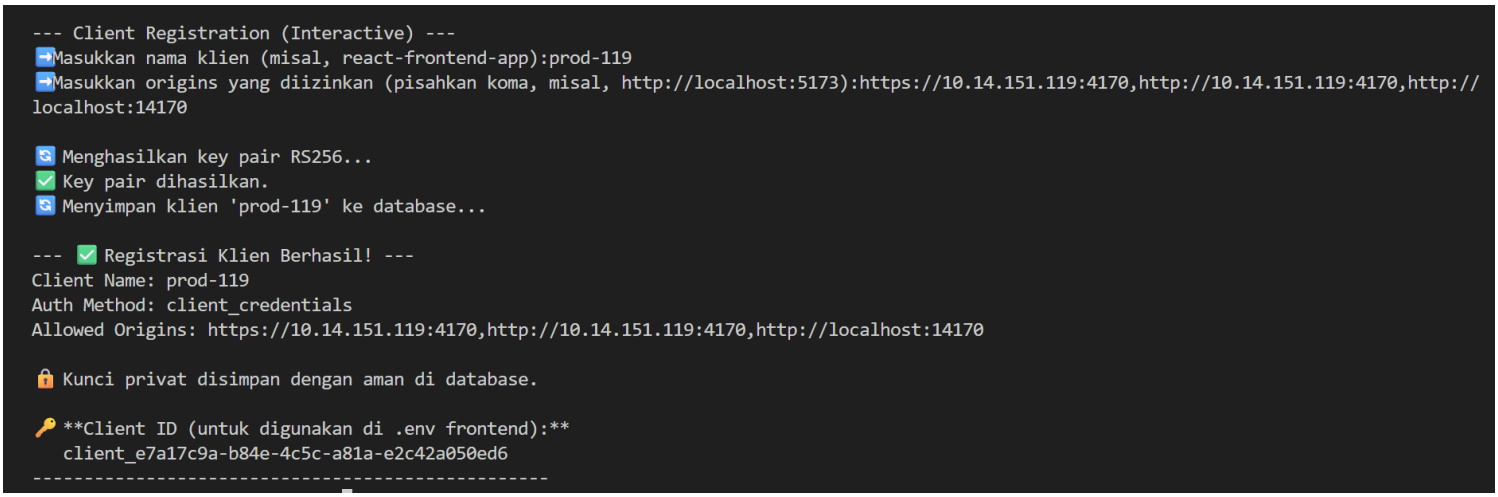
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Registrasi client dengan perintah:

- docker compose exec gateway npm run register_client

Masukkan nama klien, contoh: **prod-119**

Masukkan masukkan origins: **https://10.14.151.119:4170,http://10.14.151.119:4170, http://localhost:14170**



Salin dan simpan Client ID (client_e7a17c9a-b84e-4c5c-a81a-e2c42a050ed6) untuk digunakan nanti.

Masukkan Client ID dan Google Maps API Key:

- VITE_GOOGLE_MAPS_API_KEY=your_google_maps_key
- VITE_CLIENT_ID=your_client_id

Ubah konfigurasi di atas menjadi:
Isi dengan google maps api key anda

- VITE_GOOGLE_MAPS_API_KEY=AIzaSy-ContohAPIKeyGoogleMapsDev000

Masukkan Client ID sebelumnya

- VITE_CLIENT_ID=client_e7a17c9a-b84e-4c5c-a81a-e2c42a050ed6

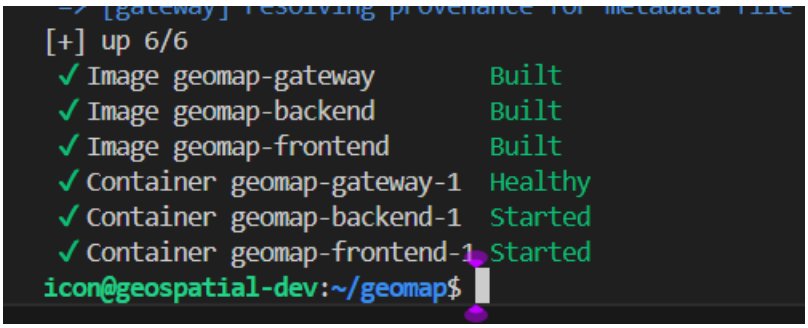
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Anda bisa melihat konfigurasi sebelum sesudahnya DI SINI

4.8. Jalankan deployment

Jalankan perintah:

```
docker compose up -d --build
```



4.9. Publish aplikasi ke internet(publik)

Salin dan Aktifkan Konfigurasi Sites-Available. Jalankan perintah:

```
# 1. Salin file dari folder proyek ke direktori sistem Nginx
sudo cp /home/icon/geomap/nginx/geomap.conf /etc/nginx/sites-available/geomap

# 2. Aktifkan konfigurasi dengan membuat symbolic link
sudo ln -sf /etc/nginx/sites-available/geomap /etc/nginx/sites-enabled/

# 3. Hapus konfigurasi default Ubuntu (agar tidak terjadi konflik port)
sudo rm -f /etc/nginx/sites-enabled/default
```

Verifikasi dan muat ulang nginx. Jalankan perintah:

```
# Verifikasi konfigurasi nginx
```

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- sudo nginx -t

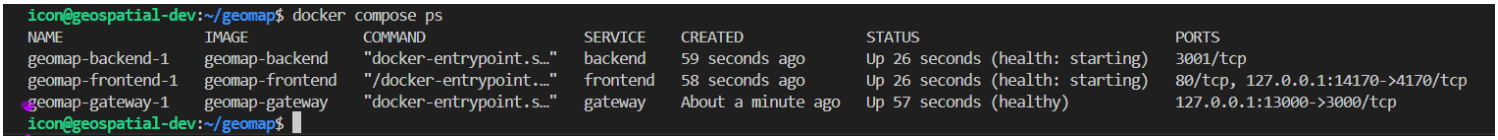
Muat ulang nginx

- sudo systemctl reload nginx

4.10. Verifikasi

Jalankan perintah:

- docker compose ps



Akses:

Frontend: https://10.14.151.119:4170

Gateway: https://10.14.151.119:3000

5. Maintenance

5.1. Pengecekan dan Pembaruan Manual SSL

Jalankan perintah:

- cd /home/icon/geomap
- ./renew_ssl.sh

5.2. Pengecekan Status Layanan

Pengecekan Layanan Nginx:

- sudo systemctl status nginx

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Pengecekan Layanan Docker:

```
sudo systemctl status docker
```

Pengecekan Layanan Geomap:

```
cd /home/icon/geomap
docker compose ps
```

5.3. Monitoring proses layanan

Monitoring Layanan Geomap:

```
cd /home/icon/geomap
docker compose stats
```

5.4. Monitoring logs

Log Geomap:

```
cd /home/icon/geomap
docker compose logs -f
# Jika ingin melihat layanan tertentu
# docker compose logs -f backend
```

6. Troubleshooting

6.1. DNS Hijacking & Kegagalan Resolusi Nama

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- **Masalah**: Muncul pesan Temporary failure in name resolution atau 404 Not Found saat menjalankan perintah apt update / apt install.
- **Penyebab**: Jaringan korporat atau ISP melakukan *DNS Hijacking* yang membelokkan permintaan DNS (seperti ke archive.ubuntu.com) menuju IP palsu.
- **Solusi**: Gunakan strategi *Static DNS*. Temukan IP mirror Ubuntu yang masih aktif (mengembalikan status HTTP 200) dengan mem-bypass DNS secara manual, lalu daftarkan IP tersebut langsung ke dalam file /etc/hosts server. Setelah didaftarkan, bersihkan *cache* paket menggunakan perintah sudo apt clean dan sudo rm -rf /var/lib/apt/lists/*, lalu jalankan kembali sudo apt update.

Jalankan perintah:

```
for ip in 185.125.190.36 185.125.190.39 91.189.91.81 91.189.91.82
  91.189.91.83 91.189.91.84; do
 echo -n "Testing $ip... "
 curl -s -o /dev/null -w "HTTP %{http_code}" --connect-timeout 5 \
  "http://$ip/ubuntu/dists/noble/InRelease" -H "Host: archive.ubuntu.com"
 echo ""
done
```

Daftarkan IP yang berhasil (status: 200):

```
echo "91.189.91.82 archive.ubuntu.com" | sudo tee -a /etc/hosts
echo "91.189.91.82 security.ubuntu.com" | sudo tee -a /etc/hosts
```

Hapus cache lama dan update metadata:

```
# Hapus cache
sudo apt clean
sudo rm -rf /var/lib/apt/lists/*

# Update metadata dari NOL
sudo apt update
```

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6.2. Akses Ditolak pada Docker (Permission Denied)

- **Masalah**: Terjadi *error* dengan pesan permission denied while trying to connect to the Docker daemon socket.
- **Penyebab**: Akun *user* yang Anda gunakan (contoh: icon) belum ditambahkan ke dalam grup keamanan docker.
- **Solusi**: Tambahkan *user* tersebut ke grup docker dengan menjalankan perintah sudo usermod -aG docker icon. **Sangat penting:** Anda wajib untuk *logout* dan *login* kembali ke dalam server agar perubahan grup tersebut diterapkan.

Jalankan perintah:

```
sudo usermod -aG docker icon

# logout dan login ulang
logout
```

6.3. APT Mengembalikan Status 404 Not Found (Metadata Usang)

- **Masalah**: Instalasi gagal dengan *error* `404 Not Found` ketika menggunakan `apt install`, sekalipun koneksi internet dan jaringan server repositori dalam keadaan normal.
- **Penyebab**: File metadata (cache) Ubuntu lokal Anda di `/var/lib/apt/lists/` belum diperbarui dan masih mencari file `.deb` versi lama yang sudah tidak tersedia lagi di server pusat.
- **Solusi**: Hapus seluruh metadata lokal yang sudah usang dengan menjalankan `sudo apt clean` dan `sudo rm -rf /var/lib/apt/lists/*`. Segera setelah itu, perbarui ulang

indeks repositori dengan menjalankan `sudo apt update`, kemudian ulangi instalasi paket Anda.

Jalankan perintah:

```
# Hapus semua cache metadata lama
sudo apt clean
sudo rm -rf /var/lib/apt/lists/*

# Download metadata segar dari repositori
sudo apt update

# Ulangi instalasi (sekarang merujuk ke versi terbaru)
sudo apt install -y <package-name>
```

7. End of Document

If you encounter problem or difficulties, you may reach our support contact at :

1. WhatsApp or GSM contact to : +62 819 4401 1272
2. Email to support@kieroteknologi.com
3. Visit kieroteknologi.com and post your problem in contact us section

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